| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 3(b) | 80 | **3** | **M2** for $\frac{400}{18}\times\frac{60\times60}{1000}$ oe<br>Or **M1** for $\frac{400}{18}$<br>or for *their* speed in m/s $\times\frac{60\times60}{1000}$<br>or for $\frac{400}{1000}$ and $\frac{18}{60\times60}$ soi |
| 4(a)(i) | $\frac{1}{11}$ oe | **1** | |
| 4(a)(ii) | $\frac{1}{110}$ oe | **2** | **M1** for $\frac{1}{11}\times\frac{1}{10}$ oe |
| 4(a)(iii) | $\frac{4}{55}$ oe | **3** | **M2** for $\left(\frac{2}{11}\times\frac{1}{10}\right)+\left(\frac{3}{11}\times\frac{2}{10}\right)$ oe<br>**or M1** for $\left(\frac{2}{11}\times\frac{1}{10}\right)$ or $\left(\frac{3}{11}\times\frac{2}{10}\right)$ seen oe |
| 4(b)(i) | $\frac{1}{165}$ oe | **2** | **M1** for $\frac{3}{11}\times\frac{2}{10}\times\frac{1}{9}$ oe |
| 4(b)(ii) | $\frac{1}{5}$ oe | **5** | **M4** for $3\left(\frac{2}{11}\times\frac{1}{10}\times\left[\frac{9}{9}\right]\right)+3\left(\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}\right)$ oe<br>or **M3** for $3\left(\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}\right)$<br>or **M2** for $3\left(\frac{2}{11}\times\frac{1}{10}\times\left[\frac{9}{9}\right]\right)$ or $\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}$ oe<br>or **M1** for $\frac{2}{11}\times\frac{1}{10}\times\left[\frac{k}{9}\right]$ where $k$ is 3, 6 or 9 |
| 4(b)(iii) | $\frac{131}{165}$ oe | **2** | **M1** for 1 – (*their* **(b)(i)** + *their* **(b)(ii)**) oe |
| 5(a)(i) | 81°<br>Angle at centre is twice angle at circumference oe | **2** | **B1** for 81° |
| 5(a)(ii) | 81°<br>Alternate segment [theorem] oe | **2** | **FT** *their* **(a)(i)**<br>**B1FT** for 81° |